Bessel filter created by cascading three stages. Each stage has different values for k_1 and k_2 as provided by Table 3-1, resulting in different resistors and capacitors being used. Need a high-pass filter? Simply swap the R and C components in the circuits (leaving R_f and R_1 alone).

This type of circuit is very common for small quantity manufacturing and R&D applications; however, serious production requires the filter to be made as an *integrated circuit*. The problem is, it is difficult to make resistors directly in silicon. The answer is the **switched capacitor filter**. Figure 3-10 illustrates its operation by comparing it to a simple RC network. If a step function is fed into an RC low-pass filter, the output rises exponentially until it matches the input. The voltage on the capacitor doesn't change instantaneously, because the resistor restricts the flow of electrical charge.

The switched capacitor filter operates by replacing the basic resistor-capacitor network with two capacitors and an electronic switch. The newly added capacitor is much smaller in value than the already existing capacitor, say, 1% of its value. The switch alternately connects the small capacitor between the input and the output at a very high frequency, typically 100 times faster than the cutoff frequency of the filter. When the switch is connected to the input, the small capacitor rapidly charges to whatever voltage is presently on the input. When the switch is connected to the output, the charge on the small capacitor is transferred to the large capacitor. In a resistor, the rate of charge transfer is determined by its resistance. In a switched capacitor circuit, the rate of charge transfer is determined by the value of the small capacitor *and* by the switching frequency. This results in a very useful feature of switched capacitor

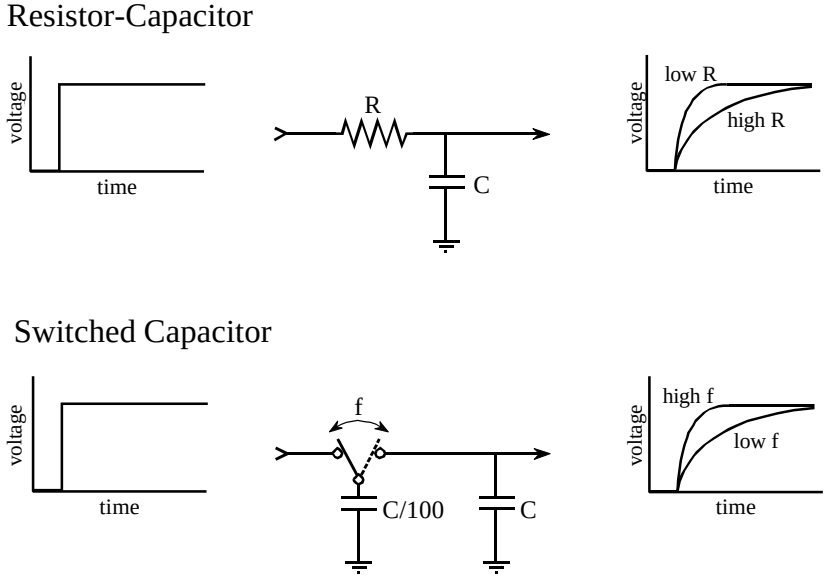

FIGURE 3-10
Switched capacitor filter operation. Switched capacitor filters use switches and capacitors to mimic resistors. As shown by the equivalent step responses, two capacitors and one switch can perform the same function as a resistor-capacitor network.